

Connect each specimen to its image and expert label

01

Register the sources

Pair each microscope image with its annotation file.

02

Map expert labels

Link original annotation titles to the target categories.

03

Audit the inventory

Check geometry, missing labels, duplicates, and slide coverage.



OUTPUT

Specimen ID + slide + coordinates + original label + mapped category

Prepare model inputs without mixing training and testing

01

Group and split

Keep related slides and every view of a specimen together.

02

Extract and inspect

Read local focal stacks; check alignment, focus, and crop margins.

03

Build model inputs

Focus-stacked tiles for detection; specimen crops for classification.

TRAINING ONLY

Augmentation and class balancing; keep evaluation sets fixed.